Jake Hoffman

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jakethehoffer.github.io/website

SUMMARY OF QUALIFICATIONS

- Computer Engineering student with a strong foundation in software engineering principles, technical problem solving, and design analysis.
- Proficient in Python, Java, C, C#, C++, SQL, Git, Word, Excel, JavaScript, Dart, Unity, FastAPI, Playwright, Monday.com, and Google Apps Script.
- Writes clean and maintainable code with attention to testing, logging, and performance.
- Collaborates with non-technical teammates to deliver features that reduce manual work and improve reliability.

EDUCATION

3rd Year at Queen's University September 2023 – Present

Computer Engineering

- Cumulative GPA of 3.85 with distinction of Dean's Scholar.
- Built a machine learning program in Python for accelerometer data that is trained to classify the difference whether the data is someone walking or jumping.
- Developed a pathfinding algorithm in Python for autonomous operation of a rover.
- Created a mobile app in Dart that pairs via Bluetooth with a "Smart Shoe"; users enter a destination and receive directions on-screen and as directional haptic feedback from the shoe.

EXPERIENCE

IT Intern Jun 2024 – Aug 2024; Jun 2025 – Aug 2025

Lynx Equity Limited

- Developed a task management system on Monday.com, streamlining communication between subsidiary companies and internal departments by automating issue routing to the appropriate workflow queue.
- Built an On/Off-Boarding workflow in Monday.com, standardizing role-based tasks, handoffs, and approvals; delivered a stakeholder demo and incorporated feedback into the final product.
- Developed Google Apps Script solutions in JavaScript that select and import data from multiple Excel exports into a single Google Sheet, apply pricing logic, and track unmatched keys for reconciliation. Added a Sheets custom menu and routines to handle large datasets, plus logic to dynamically accept changing Excel file IDs.

Game Developer (Independent) January 2024 – March 2024

Mega Tic-Tac-Toe

- Designed and programmed a Unity game built on a 3×3 grid of linked Tic-Tac-Toe boards where each move determines the opponent's next board, and winning a board claims one square on the grid.
- Supports local and online multiplayer, and single player vs. computer with multiple difficulty levels.
- Implemented a Minimax-based algorithm for the computer opponent.

Design Team Member October 2023 – March 2024

QMIND – Queen's AI Hub

- Worked on a team project to train an agent to play Blackjack using reinforcement learning.
- Built a tabular Q-learning version with a training loop and basic tests to show it worked.
- Helped the team move from the simple Q-learning version to a Deep Q Learning version so the model could learn better policies than a basic table.

Financial Analyst October 2021 – June 2023

BMC Pharmacy

- Conducted research in SW Ontario to scope animal drug compounding market opportunities.
- Tracked profitability and margin contribution for monthly/quarterly and year-over-year comparisons from drug utilization, expanded services, and other RX reports.

PERSONAL PROJECTS

trader (Python · IBKR · Finnhub · pytest)

AI swing-trading agent

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- 24/7 AI swing-trading agent for S&P 500 equities, driven by six scheduled Claude Code routines.
- Paper-traded against SPY with kill-switch, risk gates, and committed JSON journaling; graduates to live trading after 30 days of documented outperformance.
- Designed broker abstraction so the live cutover is a one-line config change (IBKR active, Alpaca preserved).

Odds Aggregator (Python · Playwright · SQLite/Alembic · FastAPI)

Production cross-book arbitrage daemon

- Production arbitrage daemon covering 10 bookmakers across 6 sports.
- Ingest → normalize → detect cross-book arbs → push alerts to Telegram and Discord. Runs 24/7 with replay tooling for postmortems.
- Built defensive ingestion: per-book health scoring, automatic backoff on bot-detection, and ten Alembic migrations from iterating on real production traffic.